

Ghost UI — Complete Product Architecture

Core Product Vision

Ghost UI is a browser-native UX intelligence engine that analyzes live interfaces, detects usability problems, and explains why interfaces succeed or fail.

Phase 1 — Foundation MVP

Goal: Detect objectively bad UX using deterministic rules.

Features include:

- Contrast detection
- Click target analysis
- Visual hierarchy checks
- Accessibility auditing
- UX scoring

Frontend Architecture

Tech Stack:

- React
- TypeScript
- Tailwind CSS
- WXT for browser extension tooling

Main components:

- Content scripts
- Overlay renderer
- Popup dashboard
- Background service worker

DOM Analysis Engine

The extension scans:

- DOM elements
- CSS styles
- Typography
- Colors
- Layout geometry
- Interactive elements

It builds an internal UI tree representation for analysis.

Rule Engine

Ghost UI uses deterministic UX heuristics rather than AI initially.

Example rules:

- Low contrast detection
- Tiny click targets
- Too many competing CTAs
- Typography inconsistency
- Excessive visual clutter

Scoring Engine

Scores are generated across categories:

- Accessibility
- Clarity
- Trust
- Interaction
- Visual hierarchy

Overlay Renderer

Issues are highlighted directly on the webpage with:

- Colored outlines
- Tooltips
- Recommendations
- Severity indicators

Phase 2 — Backend + Persistence

Backend Stack:

- Fastify
- PostgreSQL
- Prisma ORM

Features:

- Saved audits
- Team collaboration
- Public audit reports
- Historical UI comparisons

Database Design

Tables:

- users
- scans
- findings
- rules

Each scan stores findings, scores, timestamps, and issue metadata.

Phase 3 — AI Layer

AI is added after deterministic analysis is stable.

Features:

- Screenshot understanding
- AI-generated UX explanations
- Redesign recommendations
- Emotional UX analysis

Phase 4 — Advanced Intelligence

Future capabilities:

- Attention flow prediction
- Rage-click detection
- Trust scoring
- Conversion friction analysis
- Psychological UX profiling

Why This Product Is Interesting

Ghost UI combines:

- Frontend engineering
- Browser internals
- UX systems
- Accessibility
- AI-assisted interface analysis

It is technically deep, visually impressive, and highly portfolio-worthy.

Recommended MVP Timeline

Week 1:

- Extension setup
- DOM scanner

Week 2:

- Rule engine
- Basic heuristics

Week 3:

- Overlay UI
- Highlighting system

Week 4:

- Scoring engine
- Polished interface

Week 5:

- Testing
- Optimization
- Beta launch